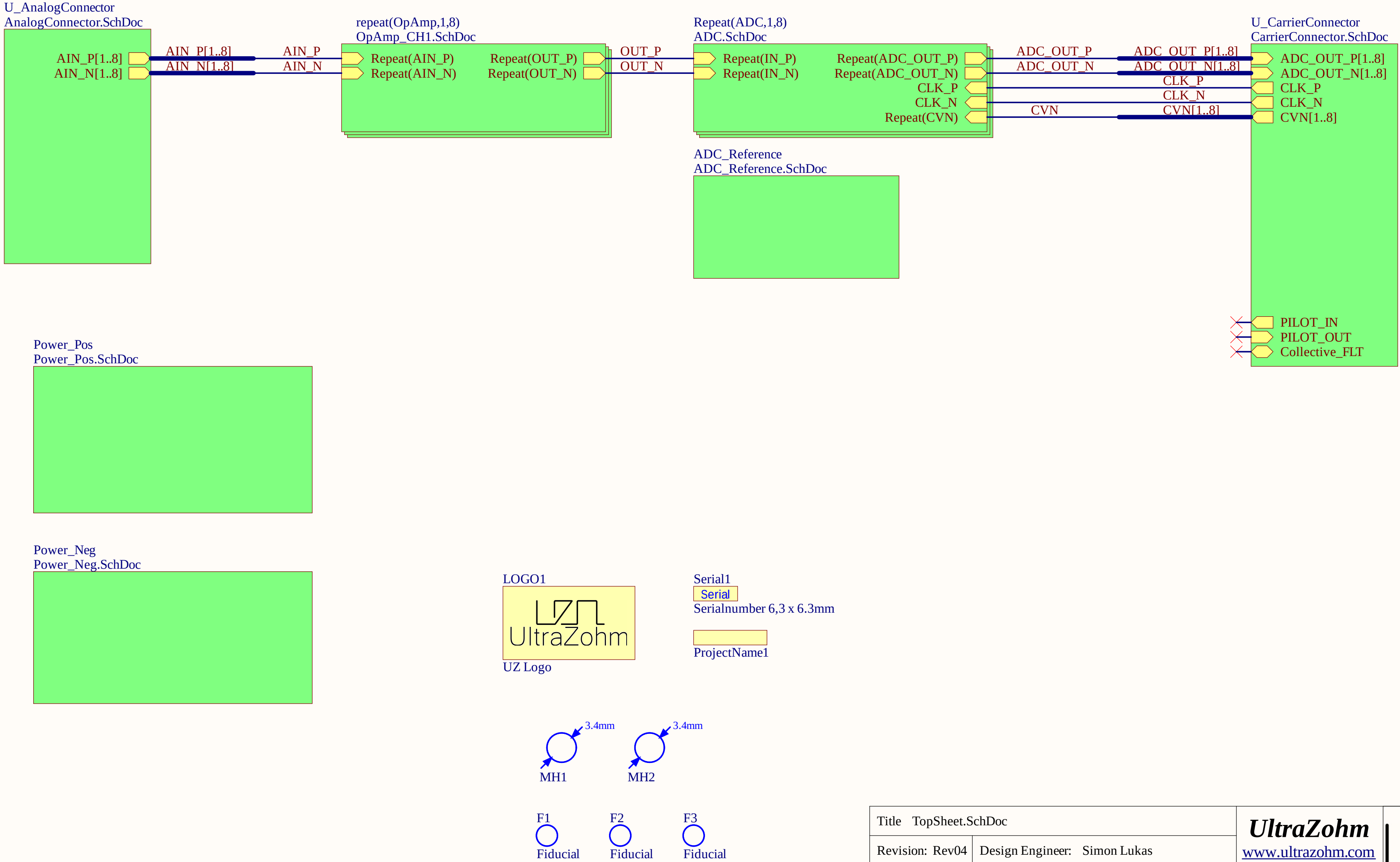
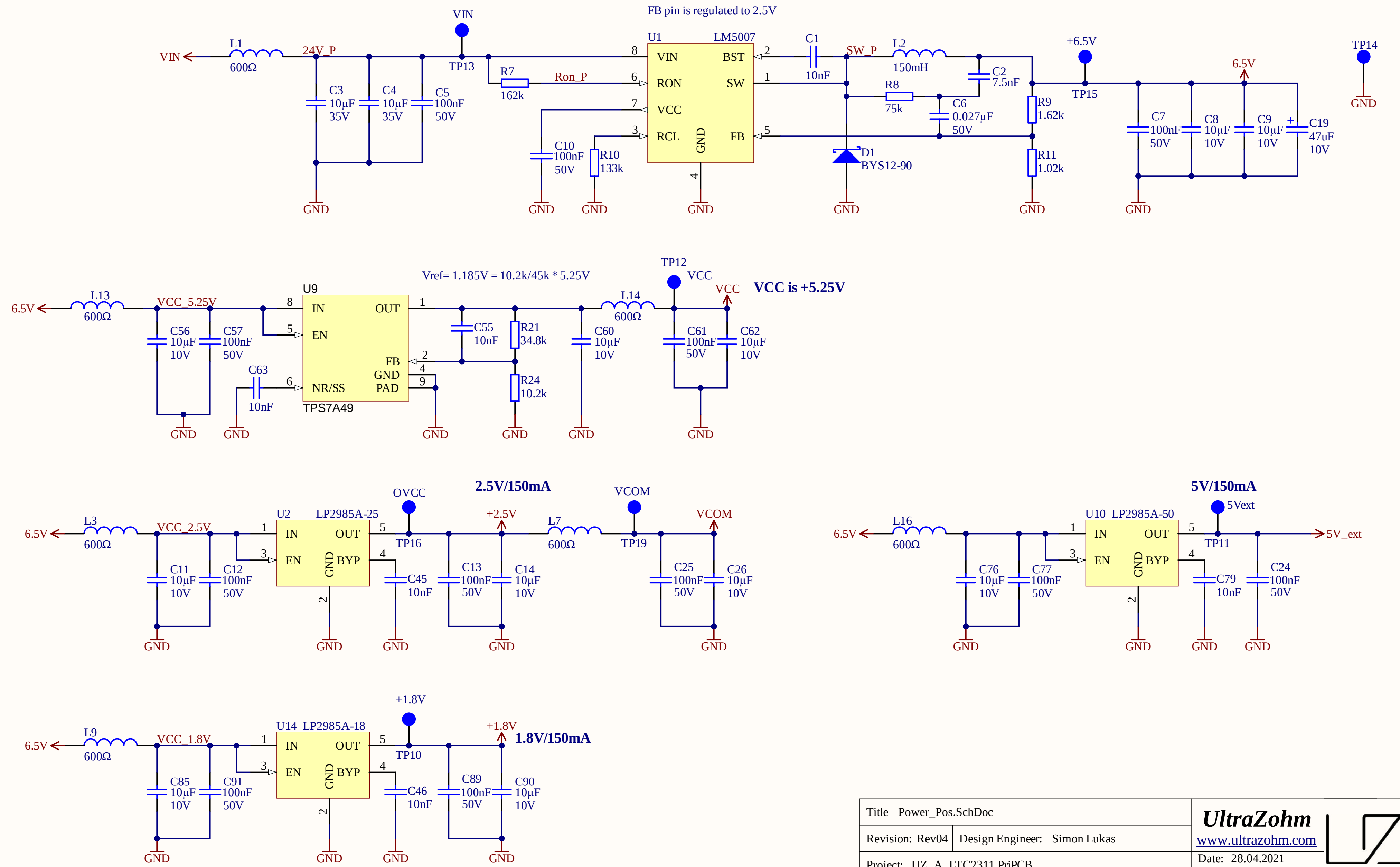
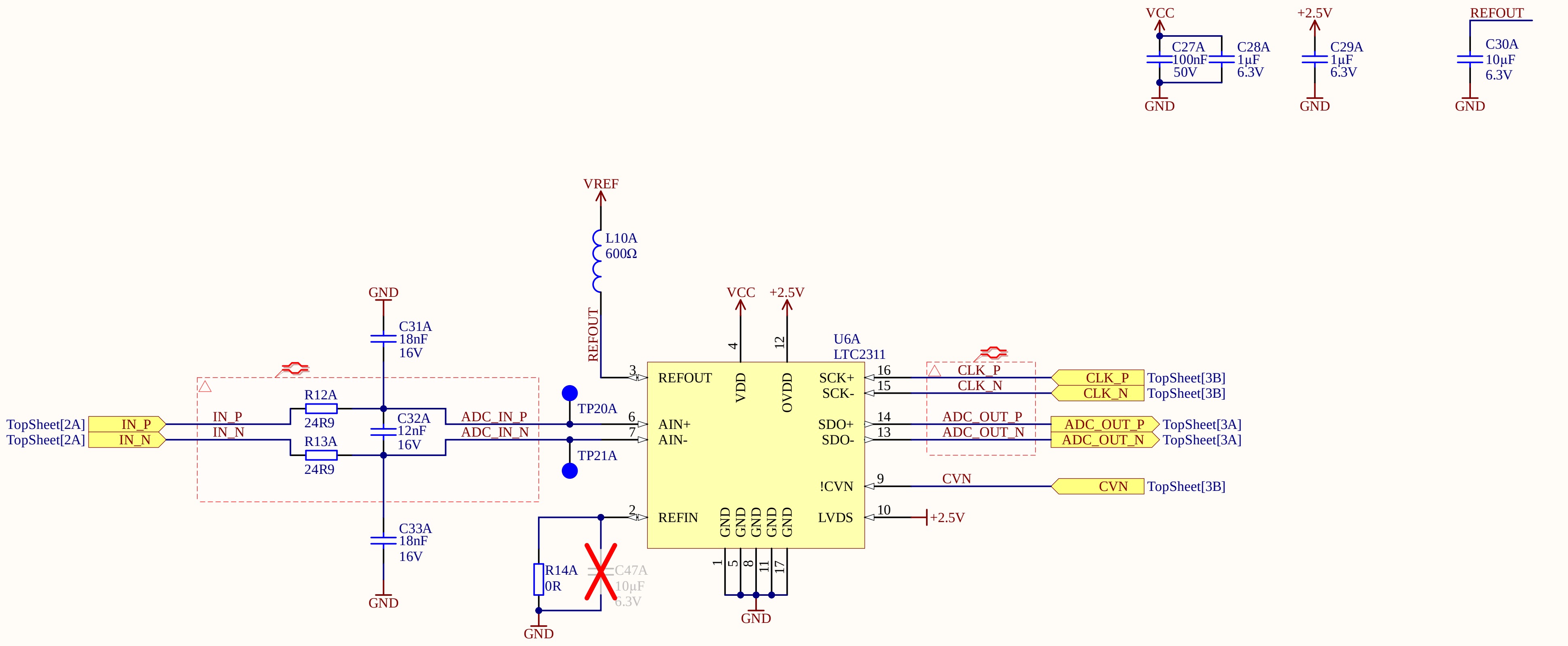


Title TopSheet.SchDoc		***UltraZohm*** www.ultrazohm.com
Revision: Rev04	Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021 Sheet 1 of 22



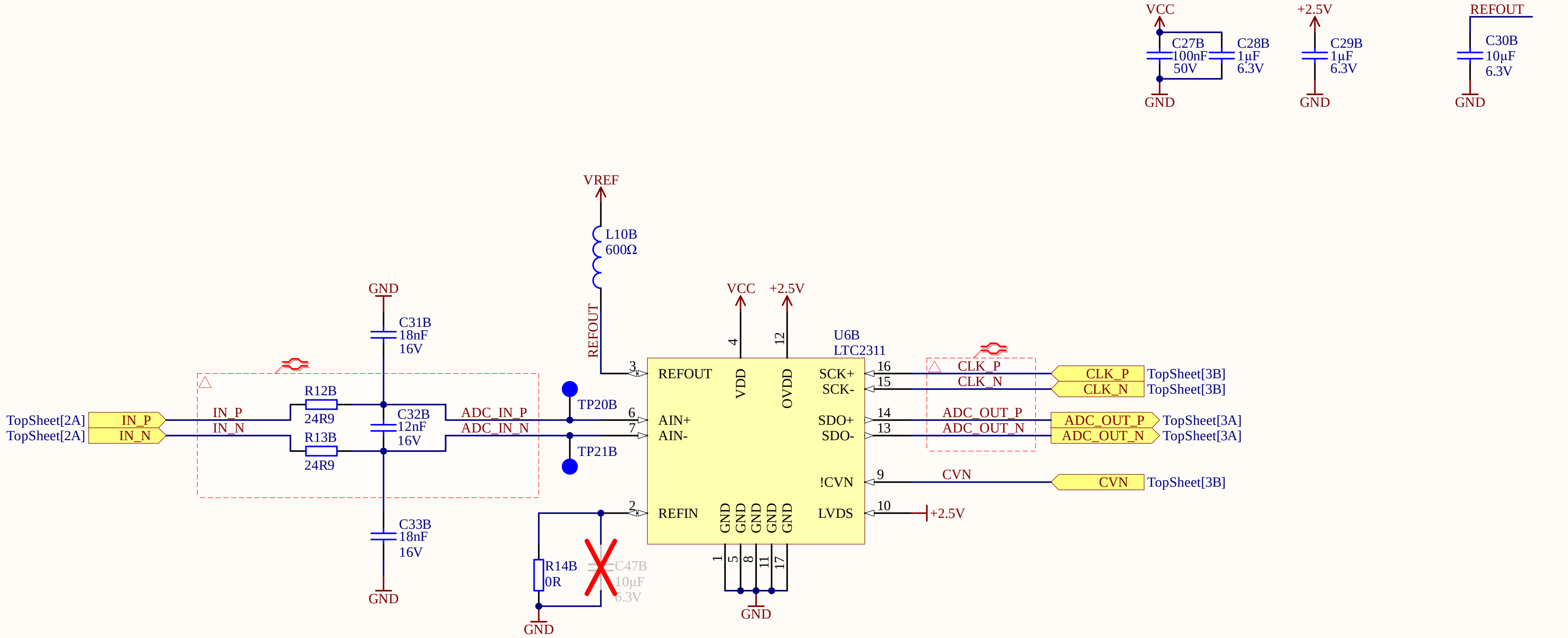
Title Power_Pos.SchDoc		UltraZohm
Revision: Rev04	Design Engineer: Simon Lukas	www.ultrazohm.com
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021
		Sheet 2 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



Title ADC.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev04	Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021 Sheet 2 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



Title ADC.SchDoc

Revision: Rev04 | Design Engineer: Simon Lukas

Project: UZ_A_LTC2311.PrjPCB

UltraZohm

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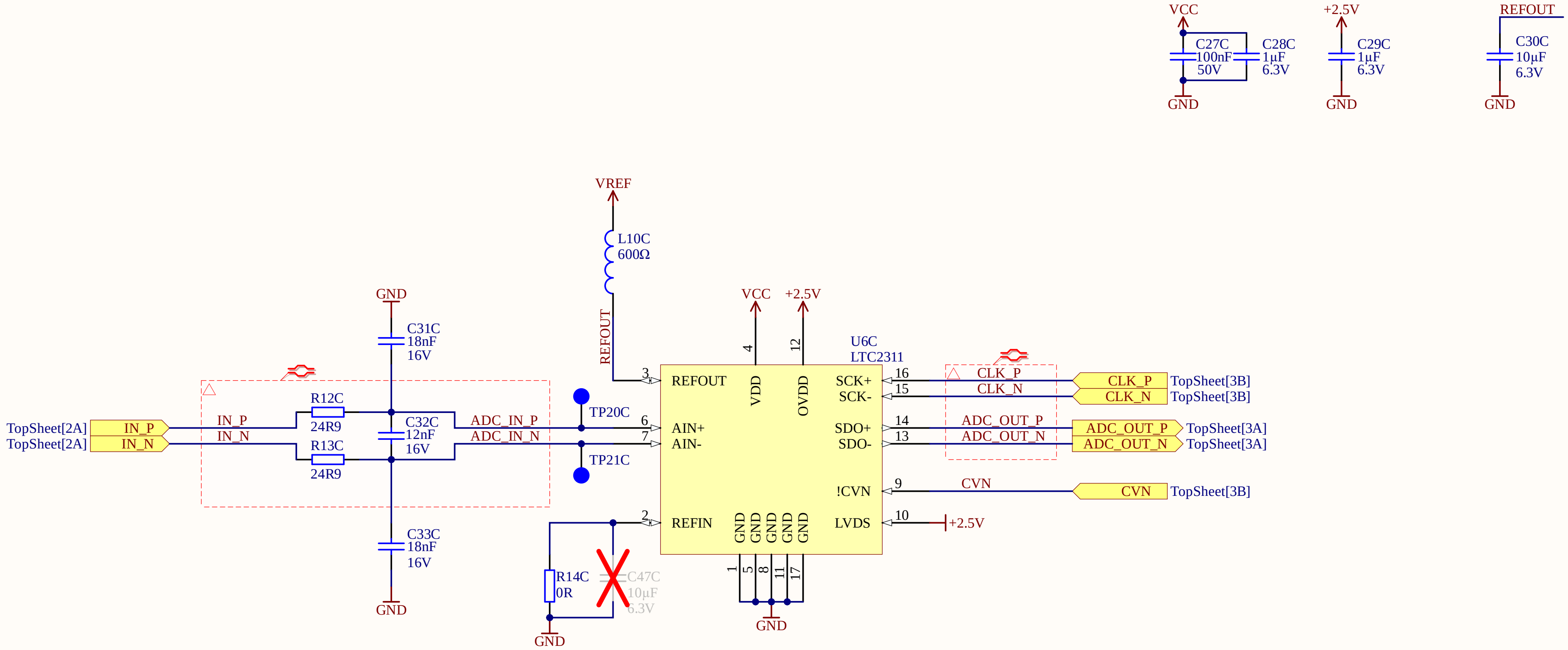
Date: 28.04.2021

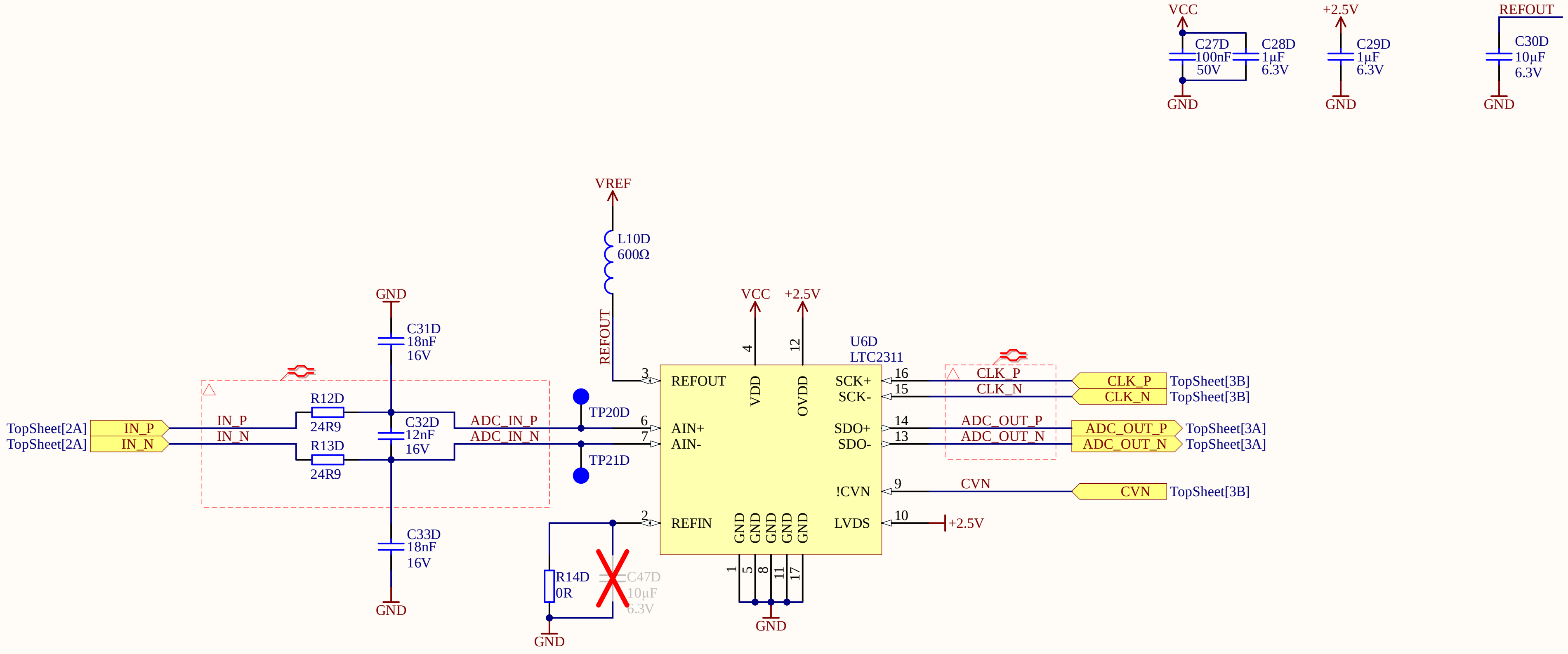
Sheet 3 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

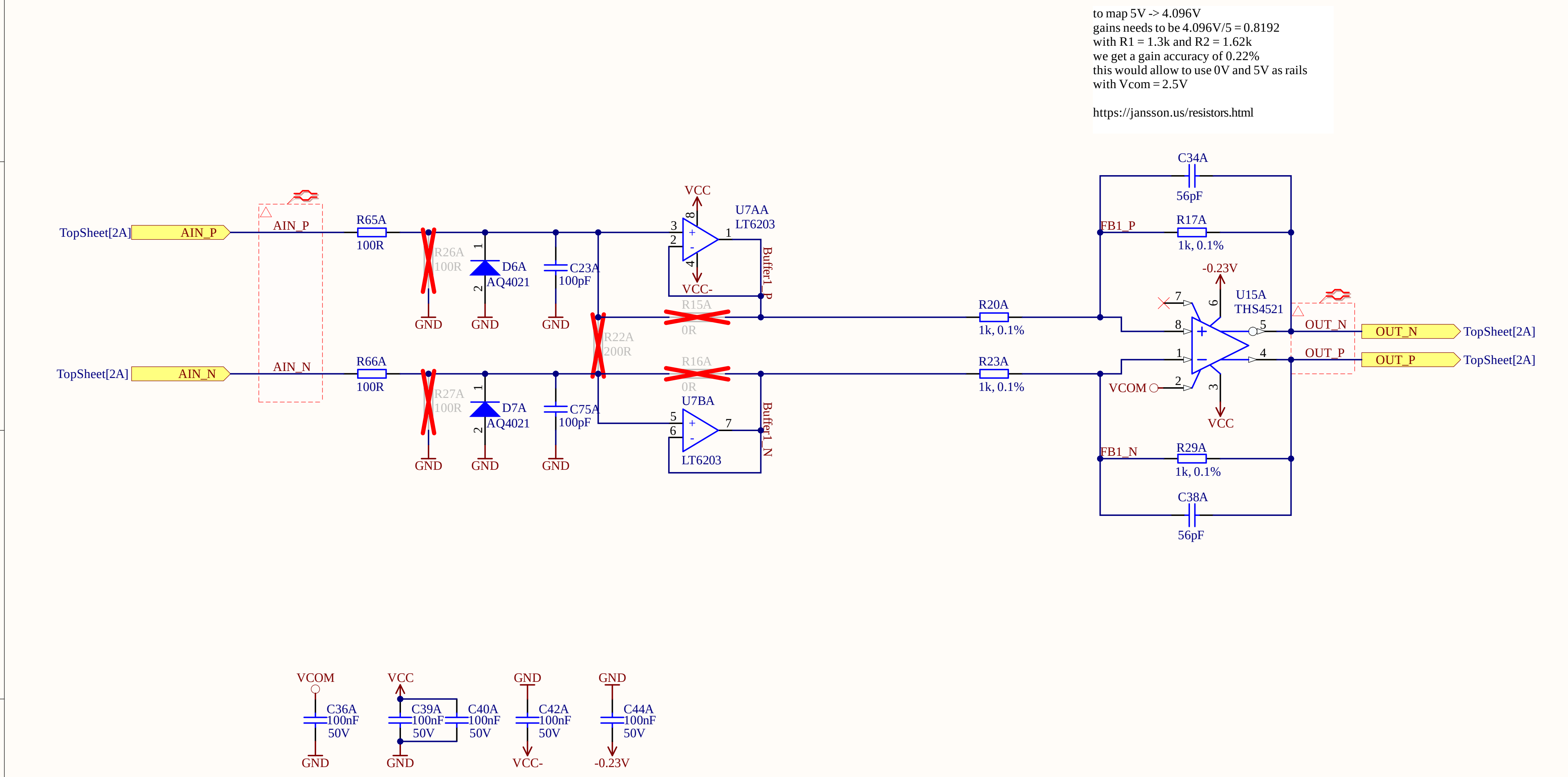


Title ADC.SchDoc		*UltraZohm* www.ultrazohm.com
Revision: Rev04	Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021 Sheet 4 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



Title ADC.SchDoc	**UltraZohm**
Revision: Rev04 Design Engineer: Simon Lukas	www.ultrazohm.com
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 5 of 22



Title	OpAmp_CH1.SchDoc
Revision: Rev04	Design Engineer: Simon Lukas
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021
	Sheet 5.1 of 22

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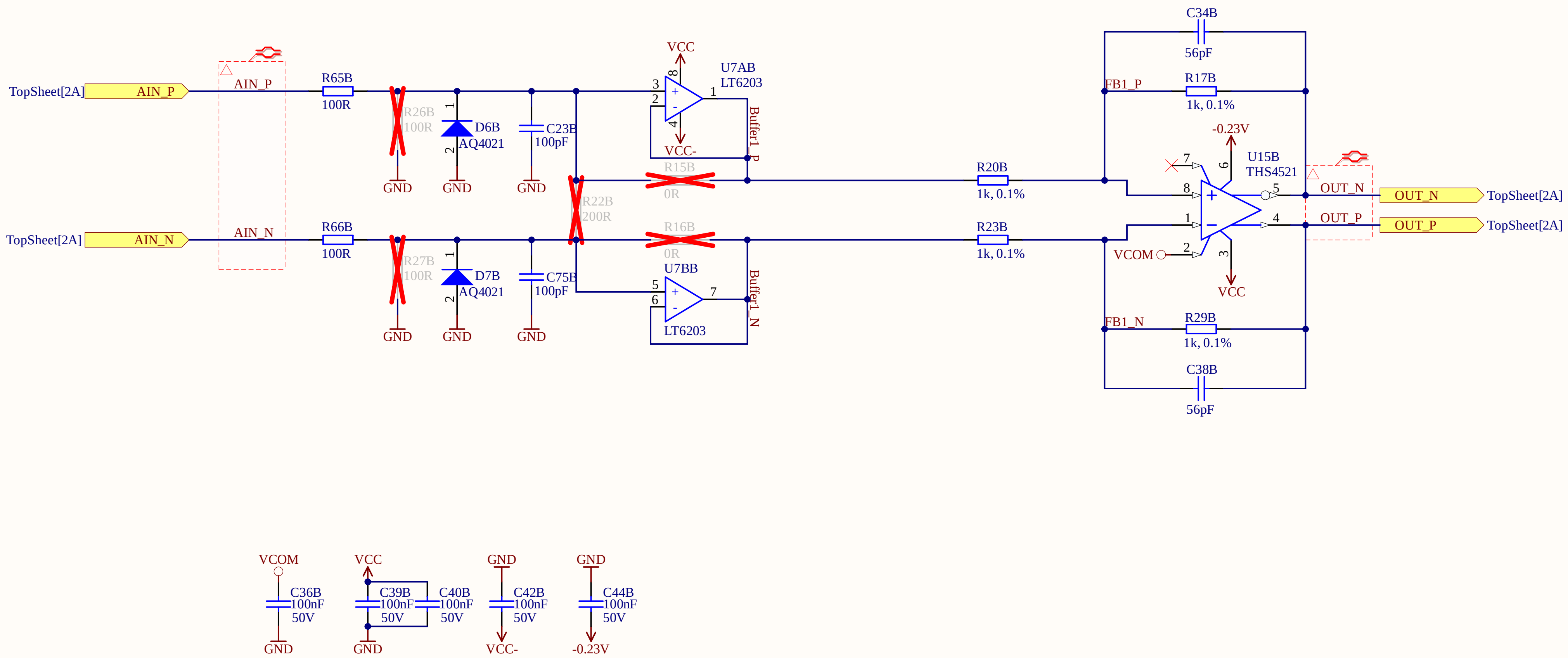
to map 5V -> 4.096V
gains needs to be 4.096V/5 = 0.8192
with R1 = 1.3k and R2 = 1.62k
we get a gain accuracy of 0.22%
this would allow to use 0V and 5V as rails
with Vcom = 2.5V

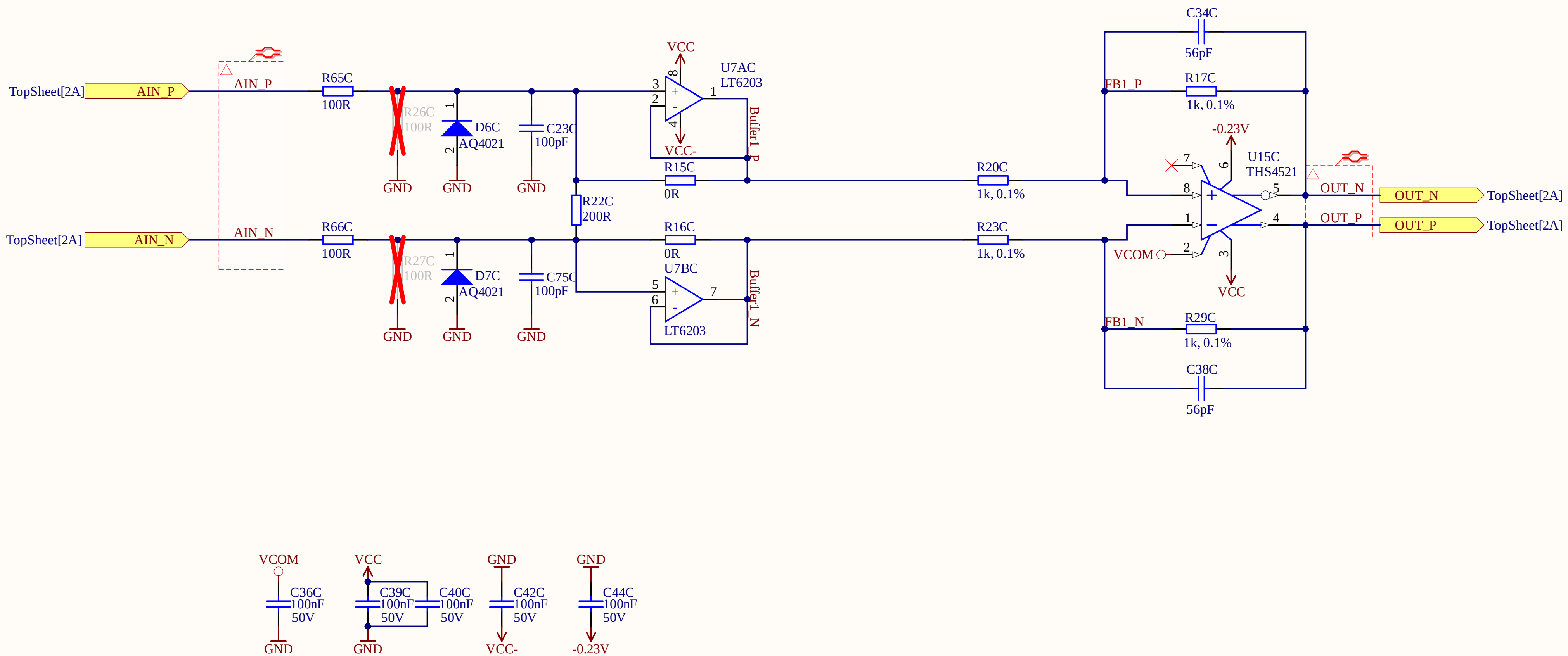
https://jansson.us/resistors.html



Title OpAmp_CH1.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04 Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 5.2 of 22

to map 5V -> 4.096V
gains needs to be 4.096V/5 = 0.8192
with R1 = 1.3k and R2 = 1.62k
we get a gain accuracy of 0.22%
this would allow to use 0V and 5V as rails
with Vcom = 2.5V

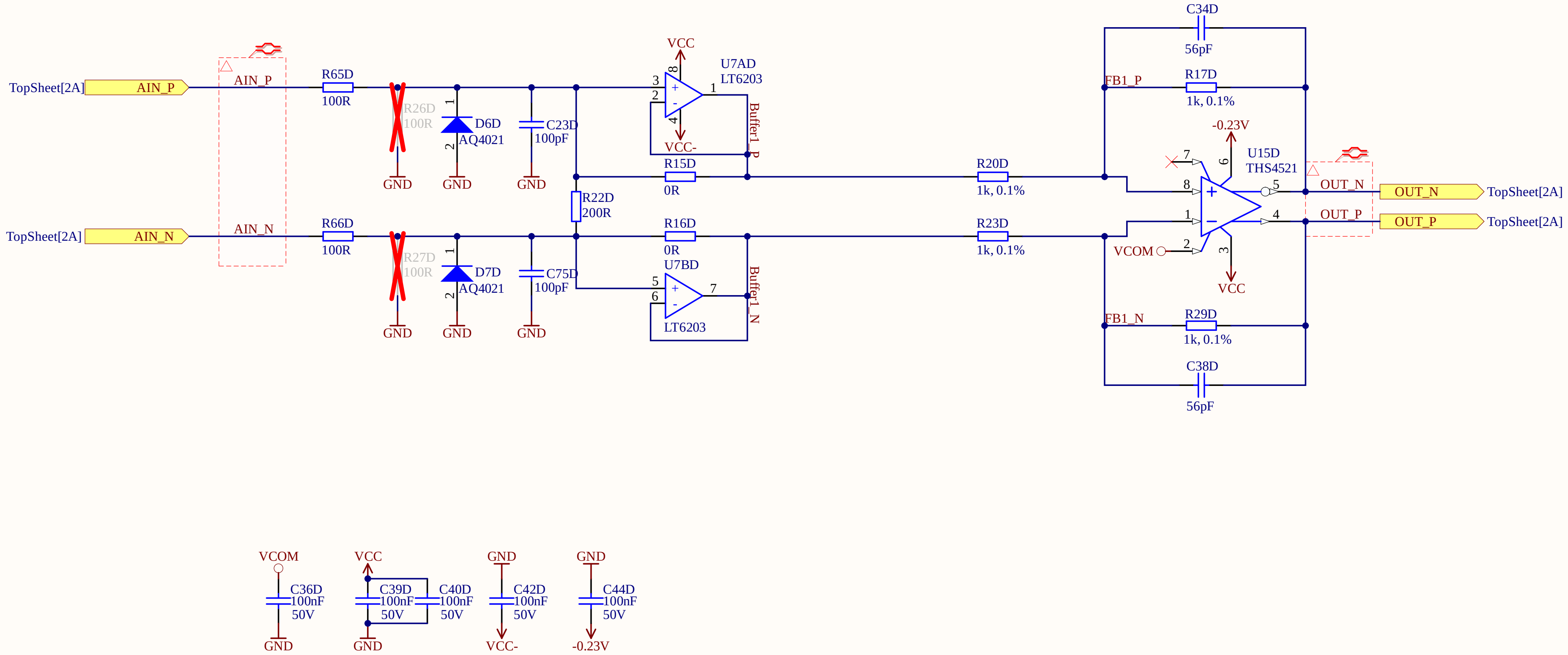
https://jansson.us/resistors.html



Title OpAmp_CH1.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04 Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 5.3 of 22

to map 5V -> 4.096V
gains needs to be 4.096V/5 = 0.8192
with R1 = 1.3k and R2 = 1.62k
we get a gain accuracy of 0.22%
this would allow to use 0V and 5V as rails
with Vcom = 2.5V

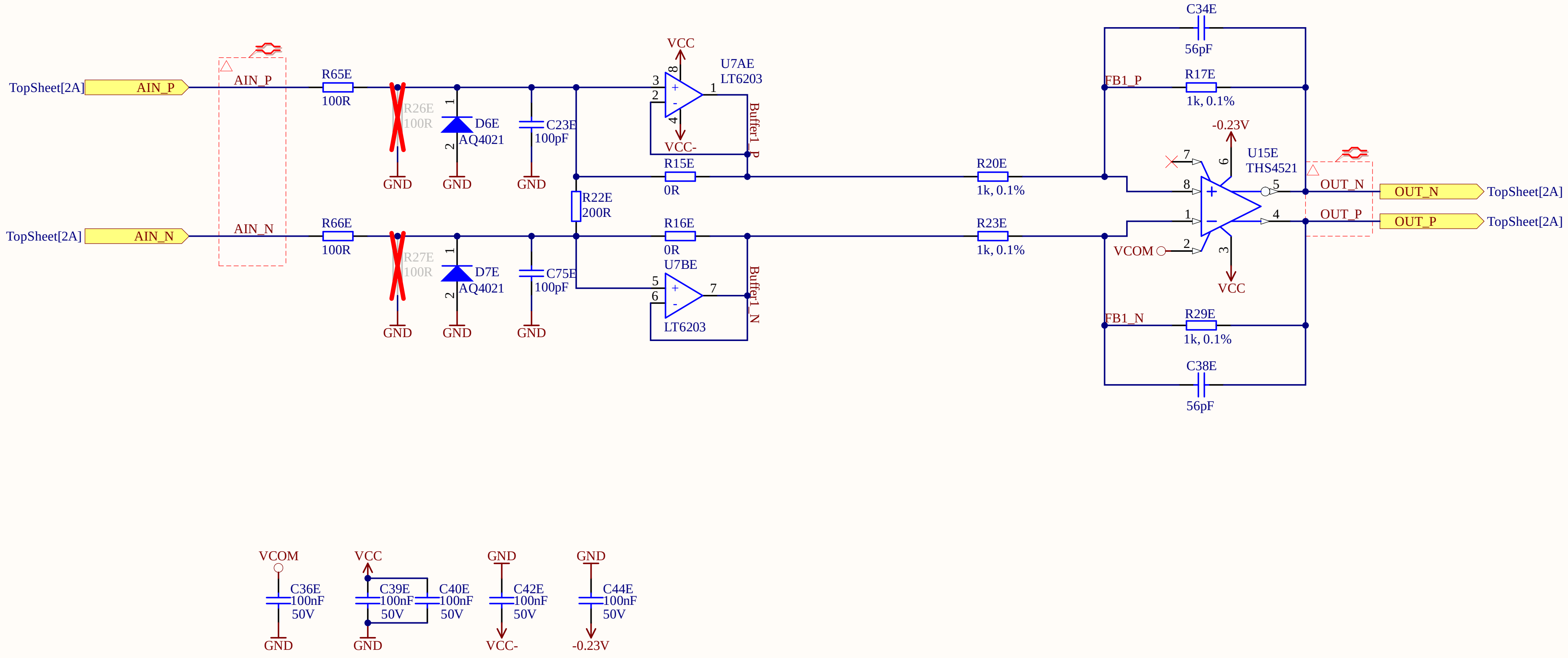
https://jansson.us/resistors.html



Title OpAmp_CH1.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev04	Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021 Sheet 5.4 of 22

to map 5V -> 4.096V
gains needs to be 4.096V/5 = 0.8192
with R1 = 1.3k and R2 = 1.62k
we get a gain accuracy of 0.22%
this would allow to use 0V and 5V as rails
with Vcom = 2.5V

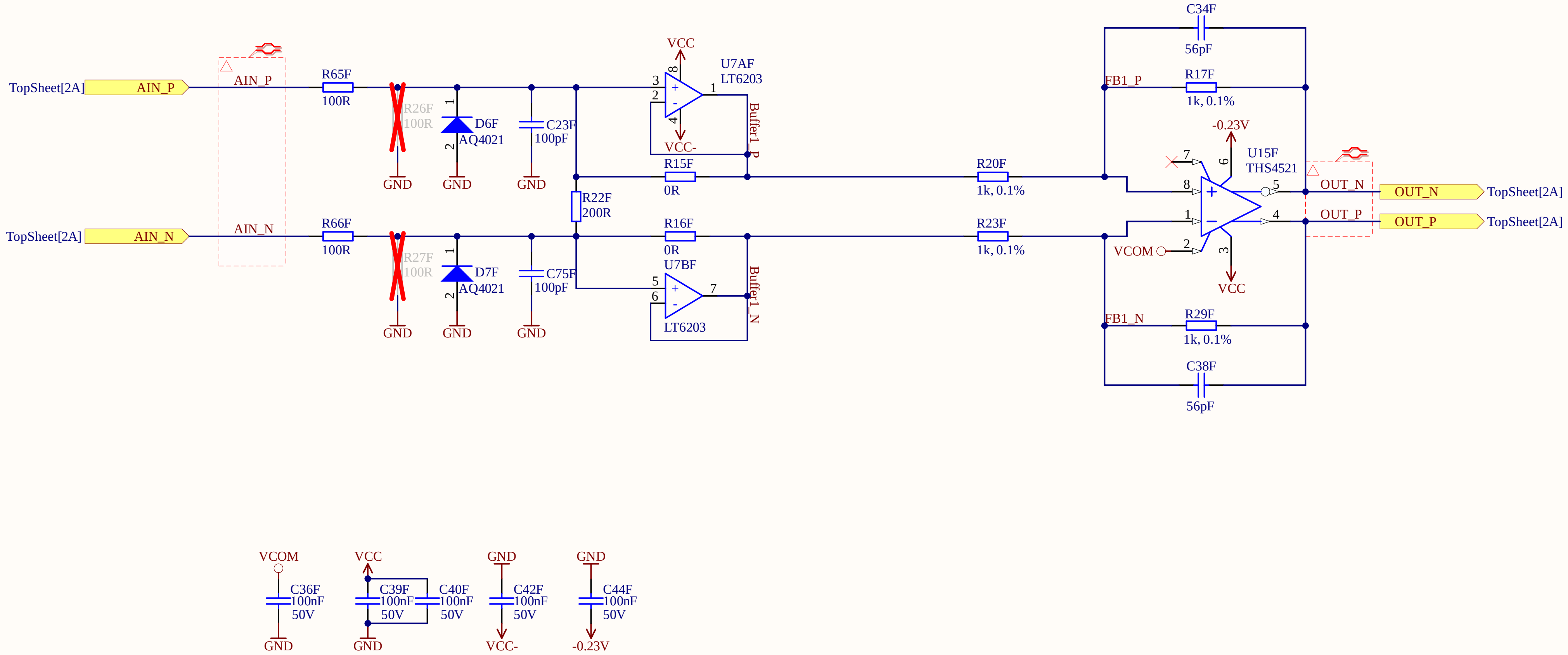
https://jansson.us/resistors.html



Title OpAmp_CH1.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04 — Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 5.5 of 22

to map 5V -> 4.096V
gains needs to be 4.096V/5 = 0.8192
with R1 = 1.3k and R2 = 1.62k
we get a gain accuracy of 0.22%
this would allow to use 0V and 5V as rails
with Vcom = 2.5V

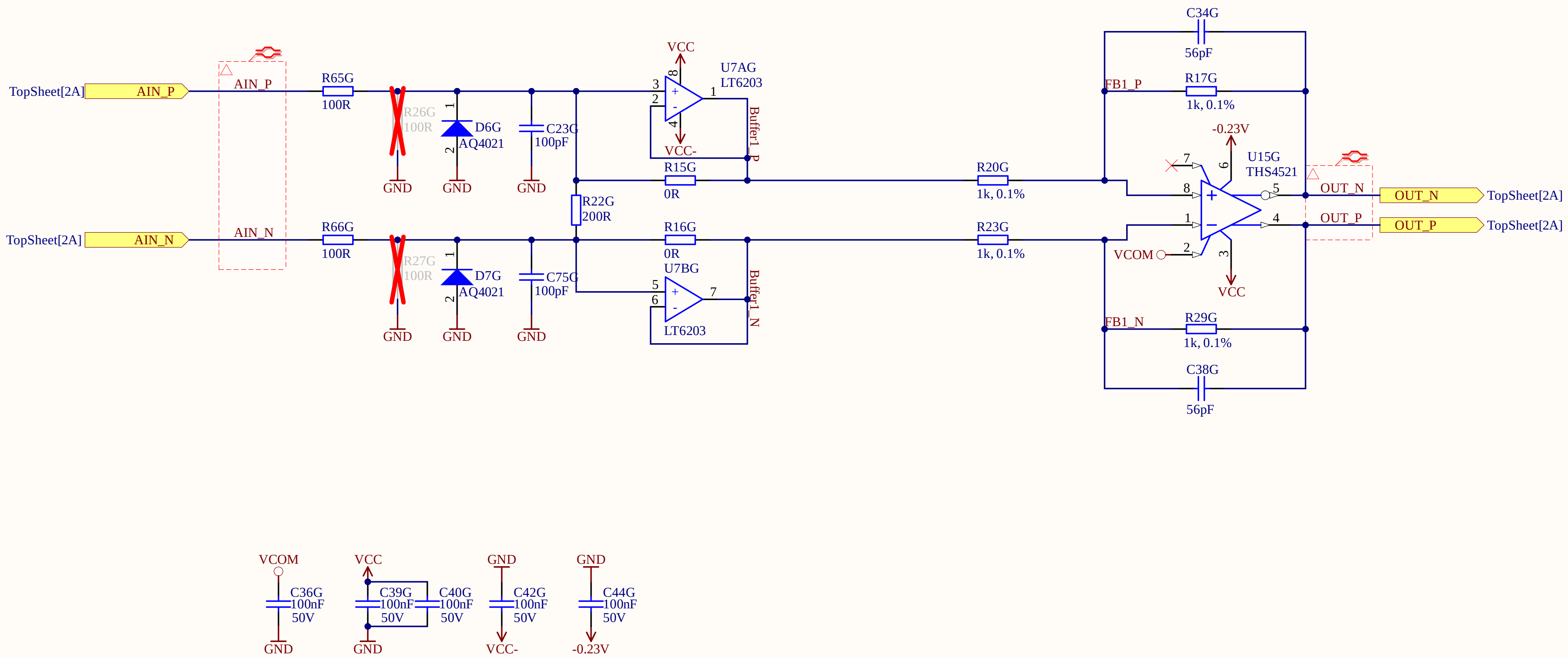
https://jansson.us/resistors.html



Title OpAmp_CH1.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04 Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 5.6 of 22

to map 5V -> 4.096V
gains needs to be 4.096V/5 = 0.8192
with R1 = 1.3k and R2 = 1.62k
we get a gain accuracy of 0.22%
this would allow to use 0V and 5V as rails
with Vcom = 2.5V

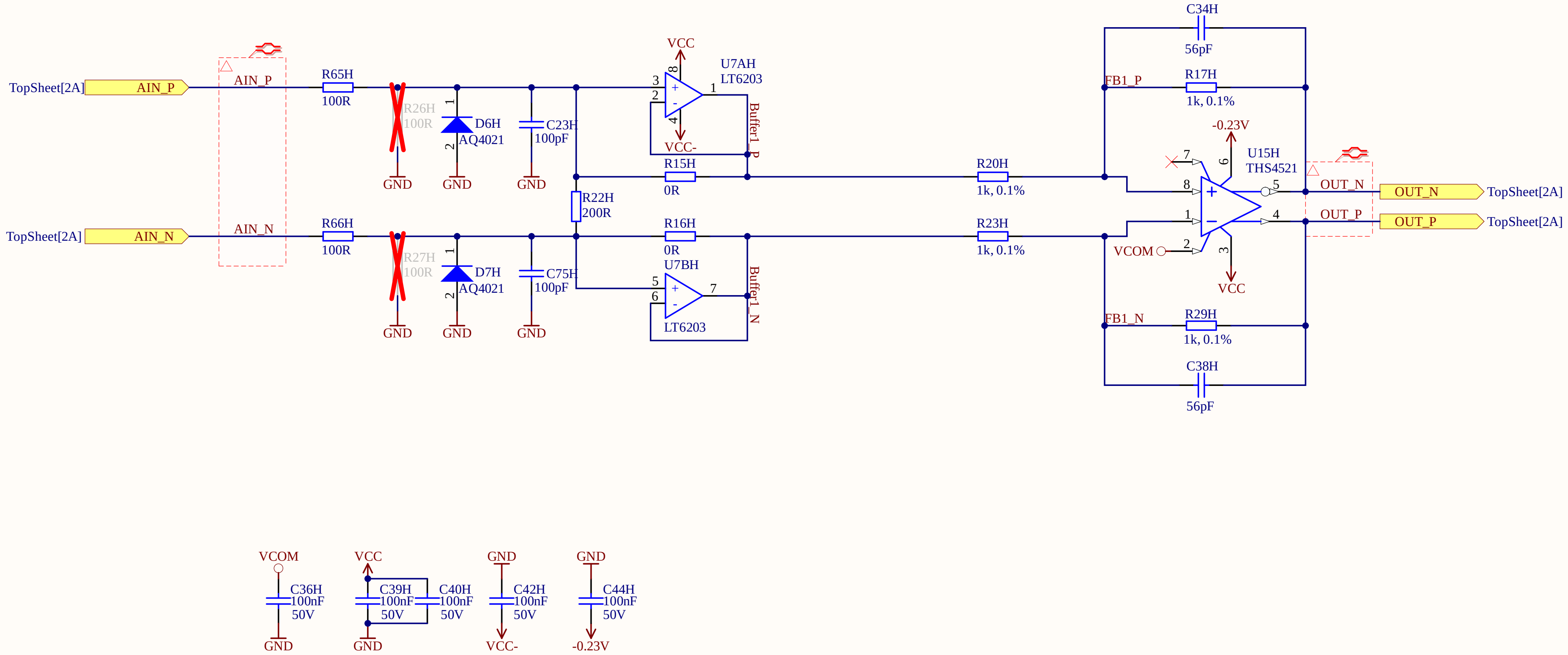
https://jansson.us/resistors.html



Title OpAmp_CH1.SchDoc	UltraZohm www.ultrazohm.com
Revision: Rev04 Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 5.7 of 22

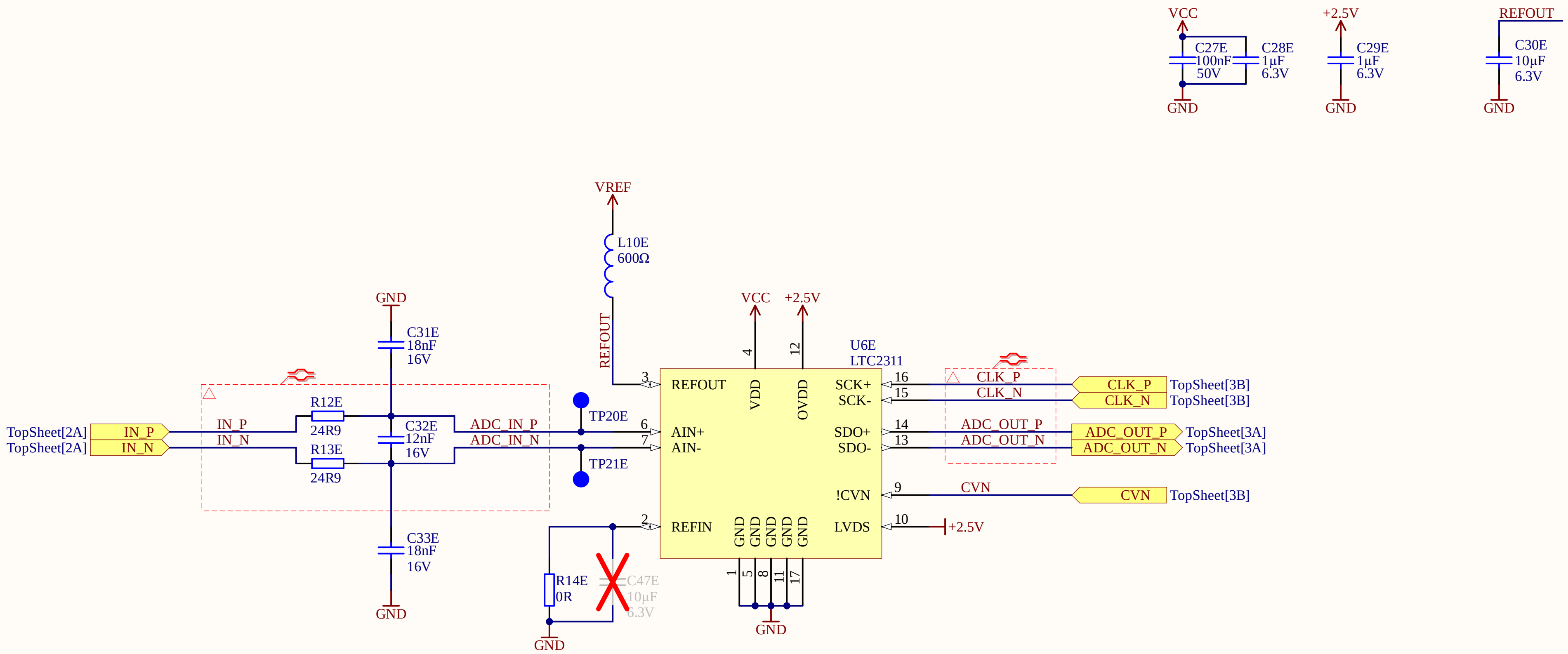
to map 5V -> 4.096V
gains needs to be 4.096V/5 = 0.8192
with R1 = 1.3k and R2 = 1.62k
we get a gain accuracy of 0.22%
this would allow to use 0V and 5V as rails
with Vcom = 2.5V

https://jansson.us/resistors.html



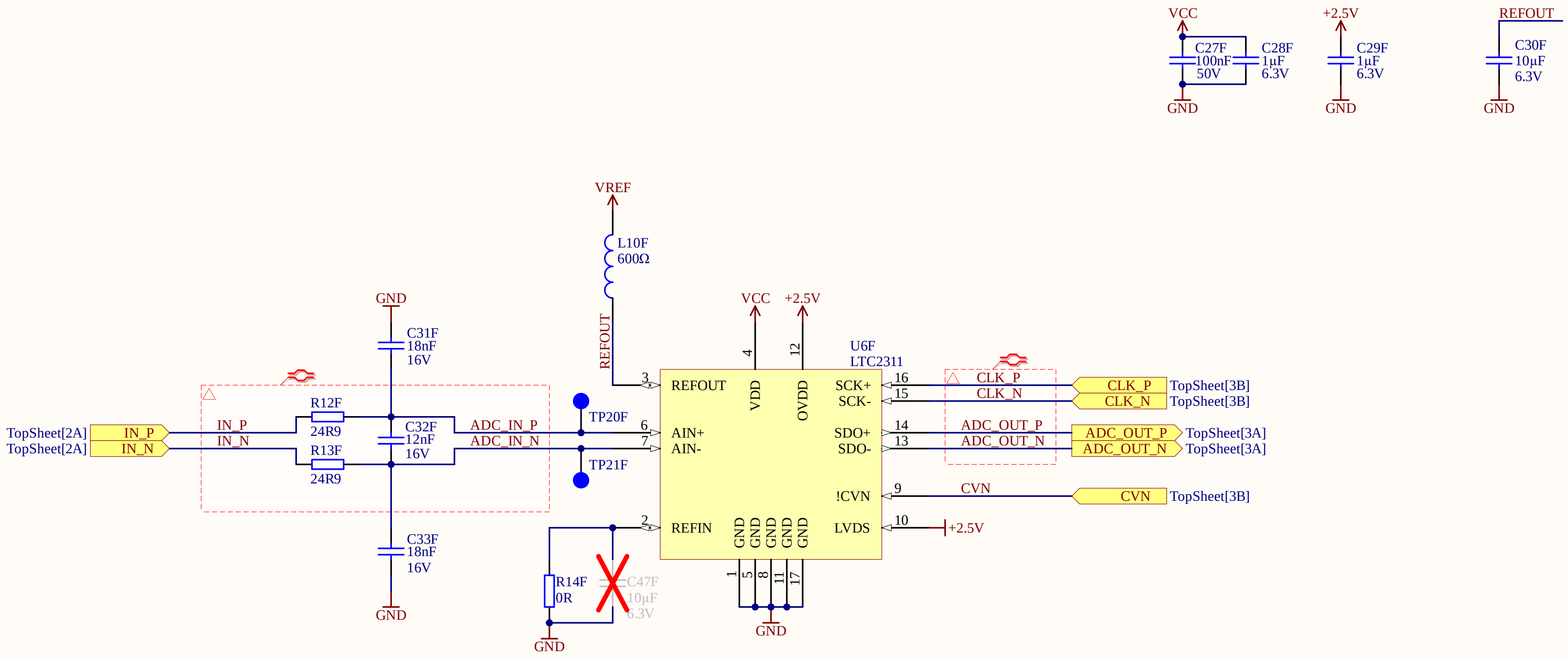
Title OpAmp_CH1.SchDoc	**UltraZohm**
Revision: Rev04 / Design Engineer: Simon Lukas	www.ultrazohm.com
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 / Sheet 5.8 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



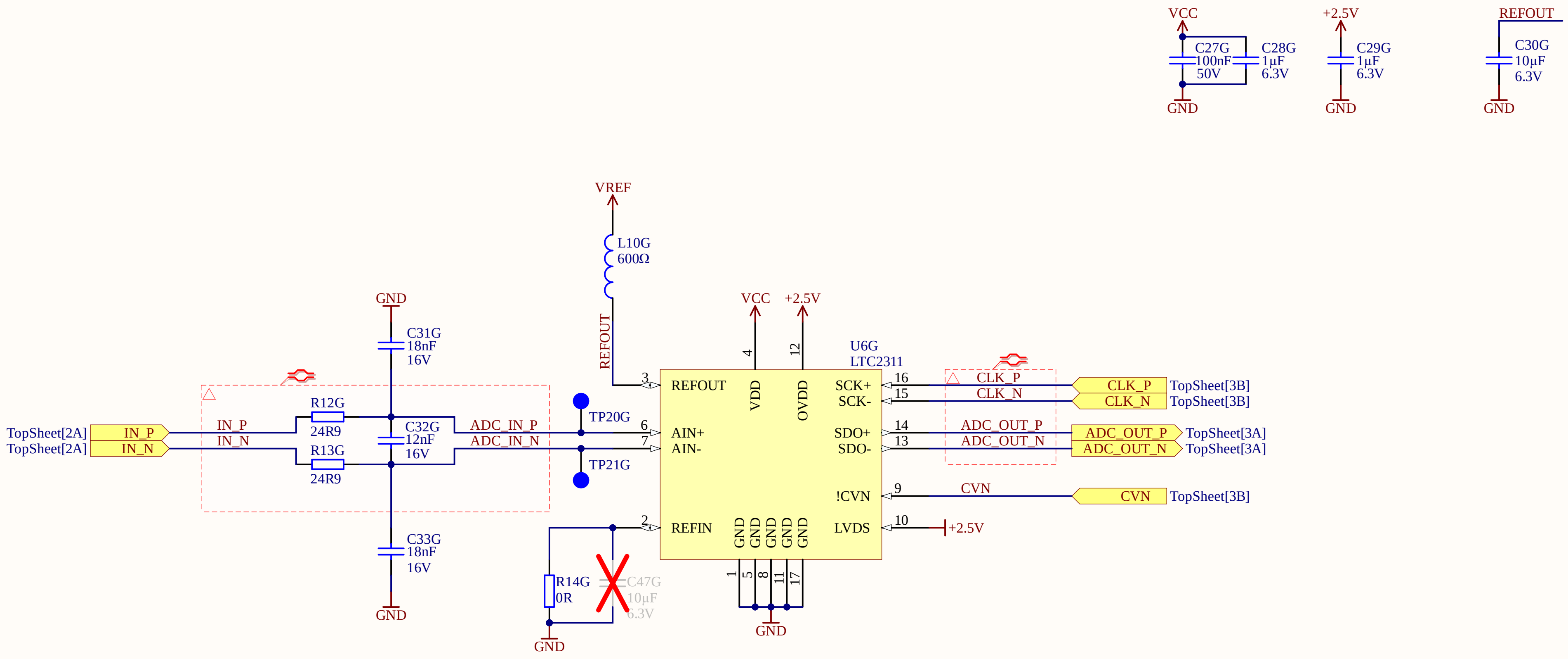
Title ADC.SchDoc		**UltraZohm**
Revision: Rev04	Design Engineer: Simon Lukas	www.ultrazohm.com
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021
		Sheet 6 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



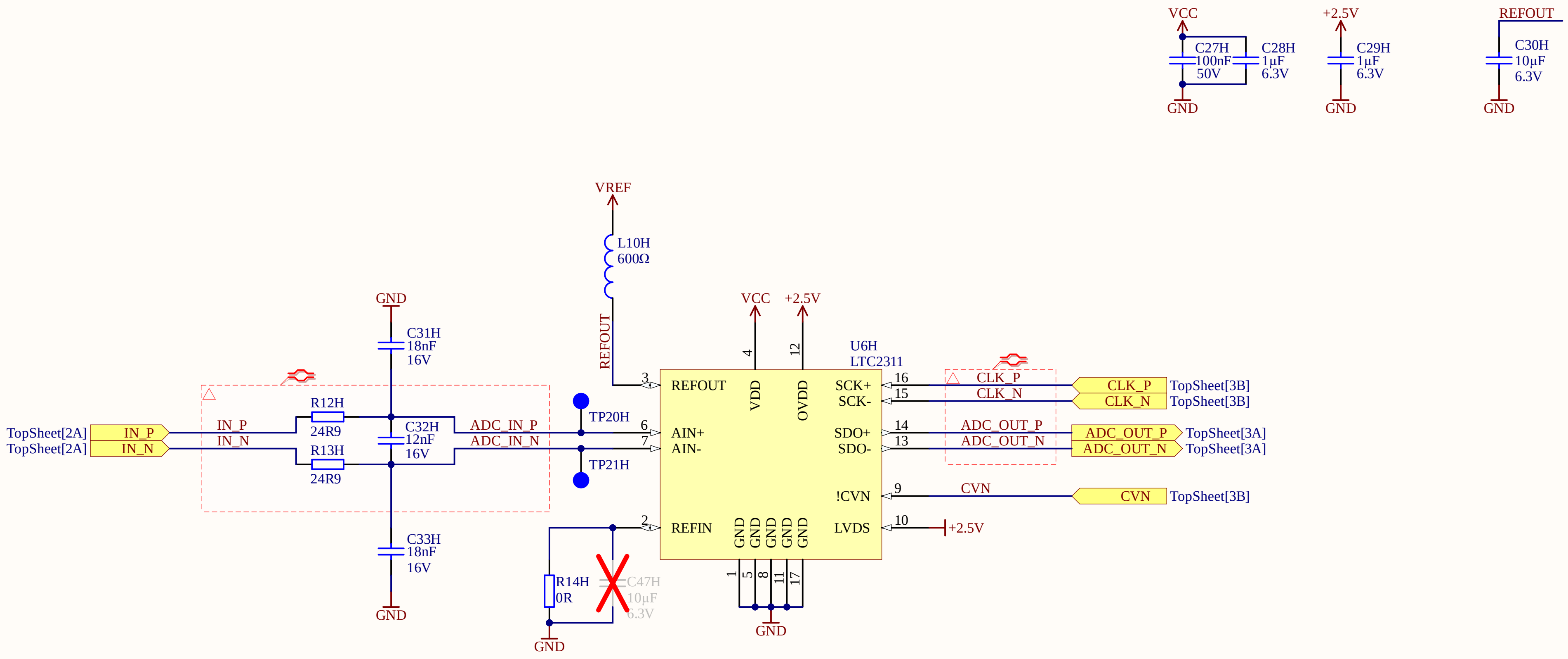
Title ADC.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev04	Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021 Sheet 7 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor

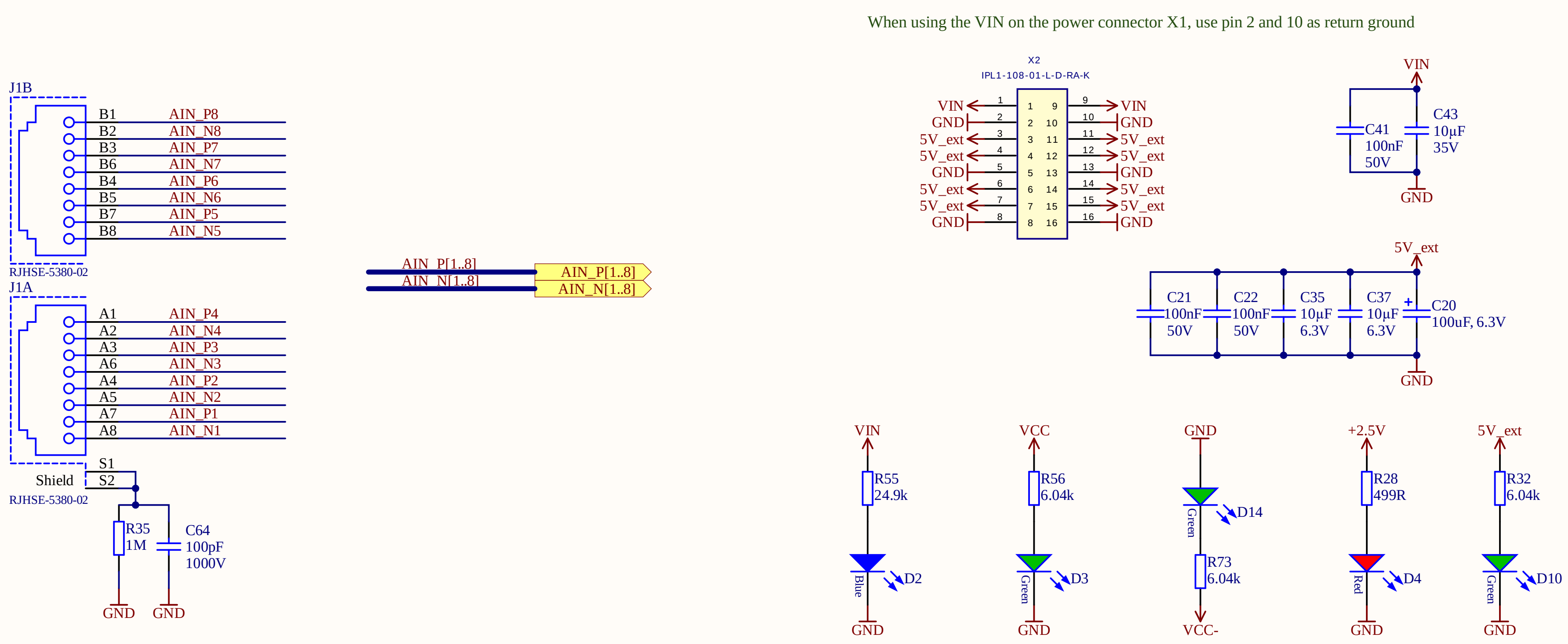


Title ADC.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04 Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 8 of 22

ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



Title ADC.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04	Design Engineer: Simon Lukas
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet 9 of 22



Title AnalogConnector.SchDoc

Revision: Rev04 Design Engineer: Simon Lukas

Project: UZ_A_LTC2311.PrjPCB

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Date: 28.04.2021

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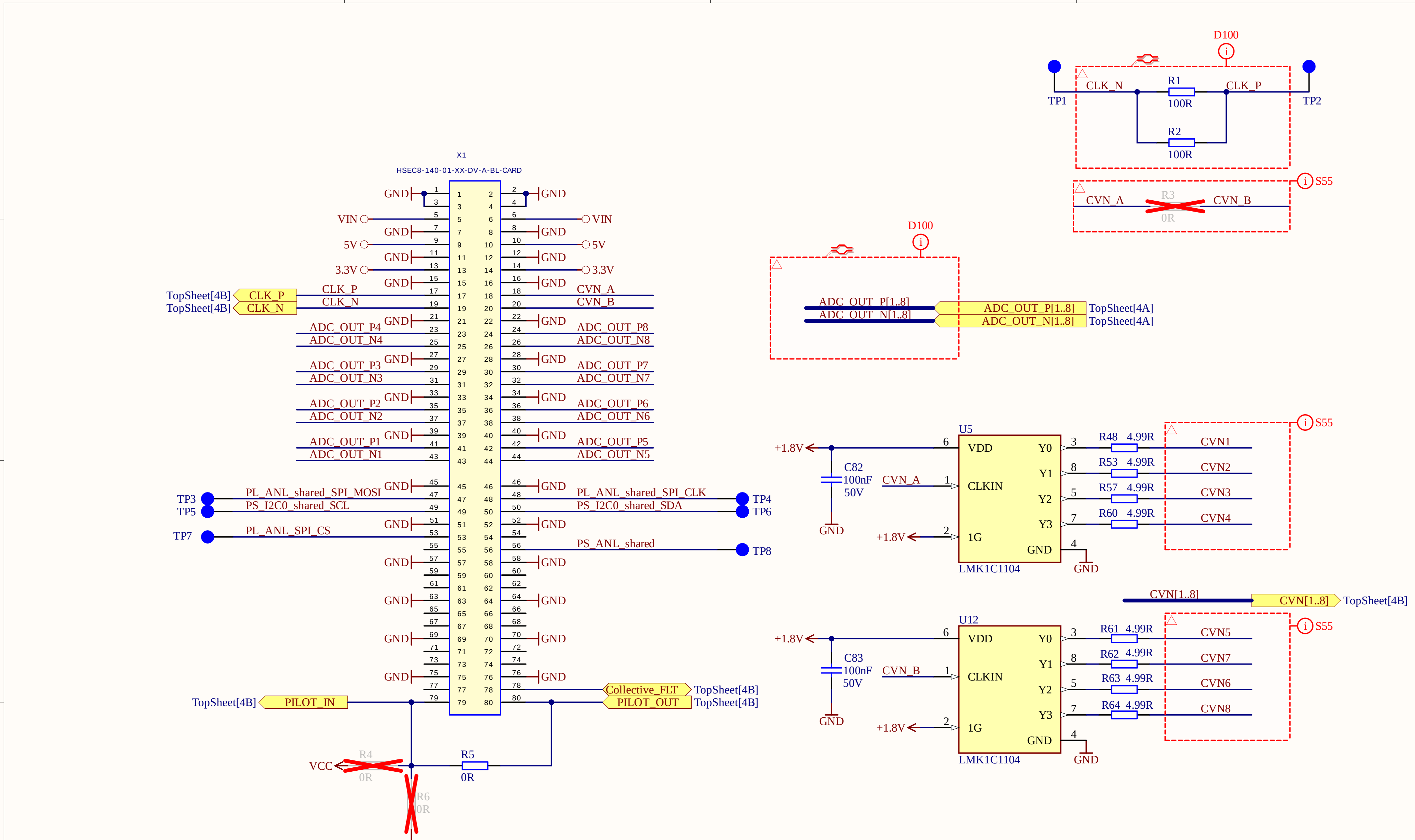
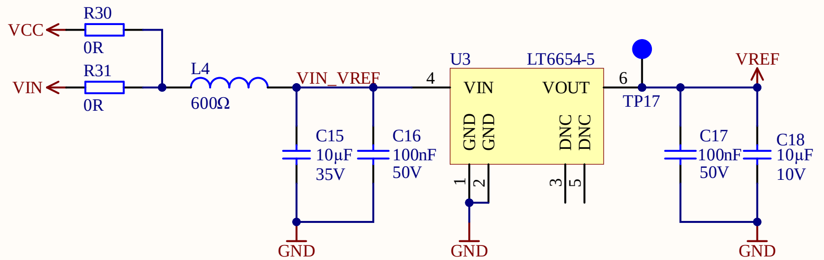
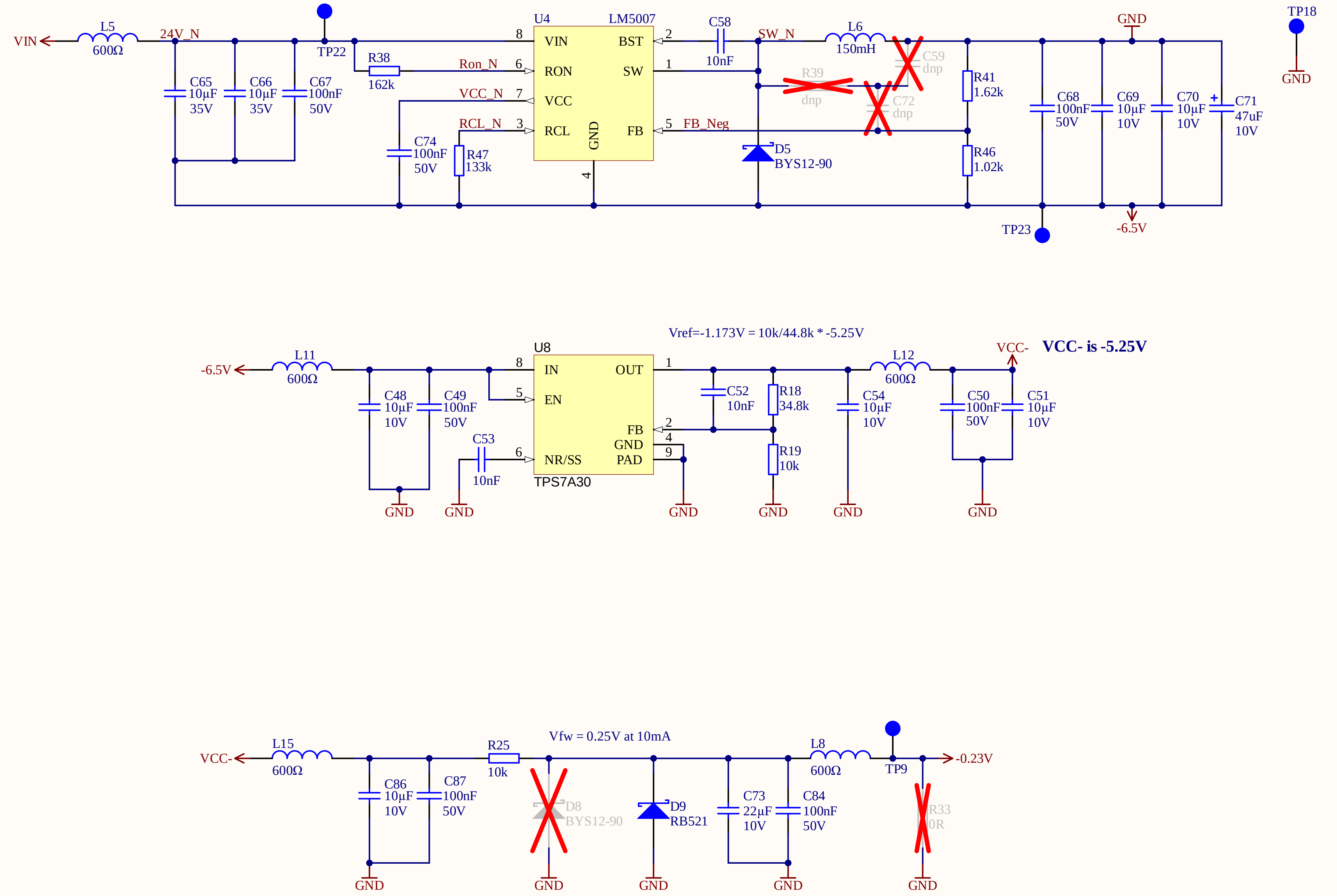


Title CarrierConnector.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev04	Design Engineer: Simon Lukas	
Project: UZ_A_LTC2311.PrjPCB		Date: 28.04.2021 Sheet 14 of 22



1. LT6654 is available in 1.25V/2.048V/2.5V/3V/3.3V/4.096V/5V
2. If the internal reference from LTC2311 (4.096V) is used, U3 is DNP, at LTC2311 REFIN needs to be connected to a cap to GND, and REFOUT disconnected
3. LTC2311 REFOUT max current sink is 700uA.
8ADCs*0.7mA = 5.6mA < 10mA max output current of LT6654

Title ADC_Reference.SchDoc	**UltraZohm**
Revision: Rev04 Design Engineer: Simon Lukas	www.ultrazohm.com
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021 Sheet of 22



Title	Power_Neg.SchDoc
Revision: Rev04	Design Engineer: Simon Lukas
Project: UZ_A_LTC2311.PrjPCB	Date: 28.04.2021
	Sheet of 22

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